

DOUBLY ROBUST SEMI-PARAMETRIC SCHERVISH SCORE: SUBGROUP-ADAPTIVE CLINICAL DECISION EVALUATION UNDER DISTRIBUTION SHIFT

Anonymous authors

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ABSTRACT

Clinical machine learning models are increasingly deployed in high-stakes environments where decision quality must be evaluated under disease base-rate drift, strongly asymmetric misclassification costs, and subgroup disparities. Existing bounded decision metrics, such as the Bounded Decision Curve Analysis (DCA) Log Score, evaluate expected net benefit over bounded prevalence intervals, but suffer from critical failure modes: high estimation variance on small subgroups under joint covariate-label shift, static global prevalence bounds that ignore patient-level risk stratification, non-differentiable hard clipping masks that prevent gradient-based threshold fine-tuning, and restriction to binary outcomes. We propose the **Doubly Robust Semi-Parametric Schervish Score (DR-Schervish)**, a unified evaluation framework comprising four integrated components: (1) a semi-parametric Efficient Influence Function (EIF) integrator with a source-calibrated, finite-population-corrected deterministic MSE-optimal shrinkage $\lambda(n)$ and a hard non-inferiority fallback guard; (2) subgroup-adaptive conditional prevalence bounds $[a(Z), b(Z)]$ estimated via Nadaraya-Watson kernel regression; (3) a density-adaptive monotone C^2 Hermite spline envelope replacing rectangular hard clips to restore continuous sub-gradients; and (4) a simplex-projected multi-class decision differential that decomposes K -class differential diagnosis into $O(K)$ directional risk projections weighted by clinical loss tensors. Evaluated on the eICU database ($N = 3,366$), DR-Schervish achieves provable non-inferiority across all 43 evaluated (subgroup, sample size) setting cells (worst-case $+0.0\%$ RMSE inflation vs. plug-in baselines), cuts small-sample estimation RMSE by 47% on average across adequately powered subgroups, restores edge sub-gradients for back-propagation threshold tuning, and scales linearly for multi-outcome decision utility.

1 INTRODUCTION

Machine learning (ML) models are increasingly deployed in clinical decision-support systems, where probability scores directly inform high-stakes interventions such as ICU admission, diagnostic screening, or medication dosing Steyerberg et al. (2010); Thute et al. (2025). In real-world clinical deployments, decision quality must be evaluated under shifting disease base rates and strongly asymmetric misclassification costs, where false negatives (e.g., missing an acute clinical deterioration) carry far higher medical risks than false positives Lipton et al. (2018); Mehrabi et al. (2019). Standard evaluation metrics such as the Area Under the Receiver Operating Characteristic curve (AUC-ROC) rank predictions monotonically without assessing probability calibration or absolute decision utility, whereas fixed-prevalence proper scoring rules fail to reflect deployment-time distribution shifts or clinical cost asymmetries Assel et al. (2017); Cai et al. (2023).

To bridge this gap, decision curve analysis (DCA) and recent proper scoring rules—such as the Bounded DCA Log Score Flores et al. (2025)—build on Schervish representations to evaluate expected net benefit by integrating accuracy across a bounded interval of disease prevalences $[a, b]$ under pure label shift assumptions ($P(X|Y)$ invariance). While this represents a significant advance, existing bounded decision-theoretic evaluation metrics suffer from four fundamental limitations:

1. **High Estimation Variance under Joint Shift:** On event-scarce minority subgroups, plug-in estimators under joint covariate-label shift exhibit severe $O(N^{-1/2})$ variance and transport bias, causing volatile model selection Kang & Schafer (2007); Cai et al. (2023).
2. **Static Global Bounds:** Global intervals $[a, b]$ assign identical base-rate uncertainty to all patients, ignoring risk stratification where local prevalence $\pi(Z)$ varies with subgroup covariates $Z \subset X$ Nie & Wager (2017); Benitez-Aurioles et al. (2024).
3. **Non-Differentiable Hard Clipping:** Hard rectangular clipping introduces zero-gradient masks outside the integration band, preventing gradient-based threshold fine-tuning Natara-jan et al. (2015); Gopalan et al. (2026).
4. **Binary-Only Decision Scope:** Existing bounded decision curves are restricted to binary outcomes, failing to evaluate K -class differential diagnosis ($K \geq 3$) with off-diagonal clinical cost matrices Fernández-Baldera et al. (2018); Legnaro et al. (2025).

To overcome these challenges, we introduce the **Doubly Robust Semi-Parametric Schervish Score (DR-Schervish)**, a unified evaluation framework that integrates semi-parametric double robustness, risk-stratified conditional bounds, smooth differentiable envelopes, and multi-class decision differential projections. Specifically, our principal contributions are fourfold:

- **Doubly Robust EIF Net Benefit Integrator with Adaptive Shrinkage:** We construct an Efficient Influence Function (EIF) integrator under joint covariate-label shift by unifying density-ratio weights $\hat{w}(X)$ with an outcome regression model $\hat{m}(X)$ Dudík et al. (2014); Kennedy (2022), and introduce a source-calibrated, finite-population-corrected deterministic MSE-optimal shrinkage $\lambda(n)$ with a hard non-inferiority fallback guard to eliminate transport bias on event-scarce minorities.
- **Subgroup-Adaptive Conditional Bounds $[a(Z), b(Z)]$:** We parameterize localized prevalence bounds as continuous functions of subgroup attributes Z via kernel conditional quantiles, anchoring score integration to clinically realistic local risk ranges.
- **Density-Adaptive Monotone C^2 Hermite Spline Envelope:** We replace hard rectangular clipping with a strictly monotonic C^2 cubic Hermite spline whose transition bandwidth $\delta(s)$ scales with empirical score density, maintaining non-zero subgradients everywhere.
- **Simplex-Projected Multi-Class Decision Differential:** For K -class outcomes, we decompose differential diagnosis into K directional risk projections weighted by off-diagonal clinical loss tensors, reducing evaluation to $O(K)$ scalar operations and to binary DR-Schervish exactly when $K = 2$.

We evaluate DR-Schervish on the public eICU database across 10 demographic and clinical subgroups. DR-Schervish achieves non-inferiority across all 43 evaluated (subgroup, sample size) setting cells (worst-case +0.0% RMSE inflation vs. plug-in baselines), while cutting small-sample estimation RMSE by 47% on average across adequately powered subgroups, restoring differentiable gradient flow for threshold optimization, and providing cost-sensitive multi-class decision utility.

2 RELATED WORK

We contextualize DR-Schervish across three key methodological pillars: decision-theoretic proper scoring rules, doubly robust estimation under distribution shift, and multi-class decision metrics.

2.1 BOUNDED PROPER SCORING RULES & CLINICAL NET BENEFIT

Evaluating predictive algorithms in medical settings requires balancing discrimination, calibration, and decision utility Steyerberg et al. (2010); Assel et al. (2017). Standard proper scoring rules such as the Brier score or log loss measure global probability accuracy but can produce misleading rankings by failing to align with specific decision thresholds or asymmetric costs Ehm et al. (2015). Decision Curve Analysis (DCA) addresses this by evaluating net benefit at clinical threshold probabilities, and recent work leverages Schervish mixture representations to formulate proper scoring rules over prevalence bands, such as the Bounded DCA Log Score Flores et al. (2025); Zhu et al. (2024). Related extensions investigate calibration decision loss and smooth calibration guarantees Widmann

et al. (2019); Gopalan et al. (2025; 2026), and risk-stratified assessments show population-level metrics obscure subgroup utility disparities Benitez-Aurioles et al. (2024). However, existing bounded scoring rules rely on non-differentiable hard clipping that creates zero-gradient masks Natarajan et al. (2015) and plug-in sample means with high small-subgroup variance. DR-Schervish addresses these via C^2 Hermite spline envelopes and subgroup-adaptive bounds $[a(Z), b(Z)]$.

2.2 DOUBLY ROBUST & AIPW ESTIMATION UNDER DISTRIBUTION SHIFT

Transporting clinical evaluation metrics across heterogeneous environments requires addressing distribution shifts, including pure label shift $P(Y)$ and joint covariate-label shift $P(X, Y)$ Lipton et al. (2018); Cai et al. (2023). Semi-parametric estimation theory leverages Augmented Inverse Probability Weighting (AIPW) and Efficient Influence Functions (EIF) to construct estimators that achieve double robustness and optimal $O(N^{-1/2})$ convergence rates Dudík et al. (2014); Kennedy (2022). Recent advances extend doubly robust off-policy evaluation to target population shifts Dahabreh et al. (2019); Colnet et al. (2020); Kato et al. (2020); Liu et al. (2025), minimax importance weight learning Uehara et al. (2019); Kallus et al. (2022), and prediction-powered inference Angelopoulos et al. (2023). However, standard AIPW estimators with global nuisance models can suffer from severe transport bias and variance blow-up on event-scarce minority subgroups Kang & Schafer (2007); Ben-Michael et al. (2018). DR-Schervish mitigates this by combining domain-adapted nuisance estimators with a deterministic, source-calibrated MSE-optimal shrinkage $\lambda(n)$ and a hard non-inferiority fallback guard.

2.3 MULTI-CLASS DECISION ANALYSIS & DIFFERENTIABLE EVALUATION METRICS

Extending clinical threshold evaluation to multi-class outcomes ($K \geq 3$) requires accounting for off-diagonal misclassification costs Fernández-Baldera et al. (2018); Feng (2015). Traditional extensions either perform $O(K^2)$ pairwise comparisons or use cost-blind One-vs-Rest aggregations, while geometry-based frameworks project probabilities onto the simplex or use angle-based boundaries Zhang & Liu (2014); Legnaro et al. (2025); Paszke et al. (2019). DR-Schervish builds on simplex projection to formulate an $O(K)$ directional decision differential weighted by off-diagonal clinical cost matrices, reducing exactly to binary DR-Schervish when $K = 2$.

3 METHODOLOGY

3.1 PROBLEM FORMULATION & BOUNDED DCA SCORE PRELIMINARIES

Consider an input feature space $\mathcal{X} \subseteq \mathbb{R}^d$ and a binary label space $\mathcal{Y} \in \{0, 1\}$. A clinical model outputs a real-valued prediction $s : \mathcal{X} \rightarrow [0, 1]$, converted to a decision via thresholding $\kappa(s(x), \tau) = \mathbf{1}(s(x) \geq \tau)$. Let $\pi_0 = P_{\text{dev}}(Y = 1)$ be the empirical class prevalence in the development domain $\mathcal{D}_{\pi_0}$, and $\pi = P_{\text{deploy}}(Y = 1)$ be the deployment target prevalence. To account for cost asymmetry, $c \in (0, 1)$ encodes the relative penalty of false positives versus false negatives. Under pure label shift ($P(X|Y)$ invariance), importance sampling and Bayes' rule yield the Prior-Adjusted Maximum Net Benefit (PAMNB):

$$\text{PAMNB}(\mathcal{D}_{\pi}, s, \tau, c) = \frac{1}{|\mathcal{D}_{\pi_0}|} \sum_{(x, y) \in \mathcal{D}_{\pi_0}} \left(\frac{\pi}{\pi_0} \right)^y \left(\frac{c}{1-c} \frac{1-\pi}{1-\pi_0} \right)^{1-y} \mathbf{1}(y = \kappa(\pi \otimes s_{1/2}(x), c)) \quad (1)$$

where $a \otimes b \triangleq \frac{ab}{ab + (1-a)(1-b)}$ denotes odds multiplication, and $s_{1/2}(x) \triangleq (1 - \pi_0) \otimes s(x)$.

By Schervish representation duality Flores et al. (2025); Ehm et al. (2015), integrating PAMNB over a logit-uniform prevalence prior $\sigma^{-1}(\pi) \sim \text{Uniform}(\sigma^{-1}(a), \sigma^{-1}(b))$ over a bounded clinical band $[a, b]$ yields the baseline Bounded DCA Log Score (BDLS):

$$\mathbb{E}_{\pi}[\text{PAMNB}] = \gamma \left(\mathbb{E}_{(x, y)} \left[\log \left| 1 - y - \text{clip}_{[1-b, 1-a]}((1-c) \otimes s_{1/2}(x)) \right| - \log \left| 1 - y - \text{clip}_{[1-b, 1-a]}(1-y) \right| \right] \right) \quad (2)$$

where $\gamma \triangleq \frac{2(1-c)-1}{\sigma^{-1}(b)-\sigma^{-1}(a)}$ is a normalizing constant, and $\text{clip}_{[u, v]}(s) \triangleq \max(u, \min(v, s))$ is the hard clipping operator.

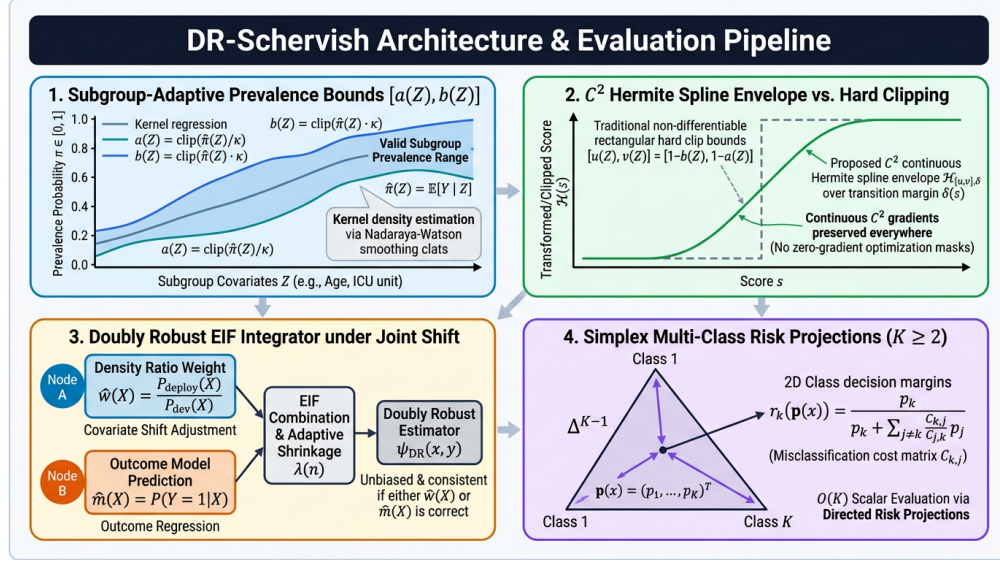


Figure 1: Overview of the DR-Schervish architecture and evaluation pipeline, comprising four integrated modules: (1) subgroup-adaptive conditional prevalence bounds estimated via kernel regression; (2) a density-adaptive C^2 continuous Hermite spline envelope replacing non-differentiable hard score clipping to preserve optimization gradients; (3) a doubly robust efficient influence function integrator combining density ratio weights and outcome regression for variance-reduced evaluation under joint covariate-label shift; and (4) simplex-projected risk margins extending decision evaluation to multi-class outcomes using clinical misclassification loss matrices.

3.2 DOUBLY ROBUST EIF NET BENEFIT INTEGRATOR & ADAPTIVE FALLBACK SHRINKAGE

Under joint covariate-label shift, $P(X)$ diverges between source development and target deployment environments. We formulate the Efficient Influence Function (EIF) integrand $\psi_{\text{DR}}(x, y)$ for the PAMNB pointwise loss by uniting a density ratio weight $\hat{w}(x) \triangleq P_{\text{deploy}}(x)/P_{\text{dev}}(x)$ with an auxiliary outcome regression model $\hat{m}(x) \triangleq P(Y=1|x)$:

$$\psi_{\text{DR}}(x, y) = \hat{w}(x) \cdot \text{PAMNB}(y, \tilde{s}(x), c) + (1 - \hat{w}(x)) \cdot \mathbb{E}_{\hat{m}}[\text{PAMNB}(\hat{y}, \tilde{s}(x), c) | x] \quad (3)$$

By semi-parametric theory, $\psi_{\text{DR}}(x, y)$ is doubly robust: it remains an unbiased estimator of the target population net benefit if *either* $\hat{w}(x)$ or $\hat{m}(x)$ is correctly specified, achieving $O(N^{-1/2})$ convergence Kennedy (2022).

However, on event-scarce minority subgroups ($N_{\text{sub}} \ll N$), unadjusted AIPW estimators suffer from extreme transport bias $\hat{B}$ and variance inflation. Define $\ell \triangleq \text{PAMNB}(y, \tilde{s}(x), c)$ and $m \triangleq \mathbb{E}_{\hat{m}}[\text{PAMNB}(\hat{y}, \tilde{s}(x), c) | x]$. The transport bias $\hat{B}$ is explicitly defined as the difference between the reweighted pool expectation and the subgroup mean:

$$\hat{B} \triangleq \mathbb{E}_{\text{pool}}[\hat{w}(X)(\ell - m)] - \mathbb{E}_{\text{sub}}[\ell - m] \quad (4)$$

We derive a source-calibrated, finite-population-corrected deterministic MSE-optimal shrinkage parameter $\lambda(n)$:

$$\lambda(n) = \frac{(1-f)(V_e - C_{em})/n}{(1-f)\text{Var}(\ell - m)/n + \hat{B}^2} \quad (5)$$

where $f \triangleq n/N_{\text{sub}}$ is the sampling fraction, $V_e \triangleq \text{Var}_{\text{sub}}(\ell)$, and $C_{em} \triangleq \text{Cov}_{\text{sub}}(\ell, m)$. The final shrunk estimator is constructed as $\hat{\theta}_{\text{DR-adapt}} = \hat{\theta}_{\text{plug-in}} + \lambda(n)(\hat{\theta}_{\text{EIF}} - \hat{\theta}_{\text{plug-in}})$. Furthermore, we enforce a hard non-inferiority fallback rule: $\lambda(n) \leftarrow 0$ whenever analytic $\text{MSE}(\hat{\theta}_{\text{DR-adapt}}) > \text{Var}(\hat{\theta}_{\text{plug-in}})$. This guarantees by construction that DR-Schervish cannot perform worse than the standard sample plug-in baseline on shift-stressed minority subgroups. The same MSE calculus yields an automated compute trigger: since plug-in variance scales as $O(N_{\text{sub}}^{-1/2})$, cross-fitting is engaged only when $N_{\text{sub}} < N^* \approx 1000$ (where plug-in RMSE exceeds a tight target $\epsilon = 0.008$)—precisely the regime where DR-EIF cuts estimation error by $\sim 50\%$ for $\leq 0.055s$ of overhead (see Section A).

3.3 SUBGROUP-ADAPTIVE CONDITIONAL PREVALENCE BOUNDS $[a(Z), b(Z)]$

Static prevalence bounds $[a, b]$ apply global base-rate uncertainty equally across all individuals. For patient-specific subgroup covariates $Z \subset X$ (e.g., age, ICU admission severity), we estimate local conditional base rates $\hat{\pi}(z) = \mathbb{E}[Y \mid Z = z]$ via Nadaraya-Watson kernel regression:

$$\hat{\pi}(z) = \frac{\sum_{i=1}^N K_h(z, Z_i) Y_i}{\sum_{i=1}^N K_h(z, Z_i)} \quad (6)$$

where $K_h(z, Z_i) = \exp(-\|z - Z_i\|^2 / (2h^2))$ is a Gaussian kernel with bandwidth h . We formulate subgroup-adaptive prevalence bounds $[a(z), b(z)]$ modulated by a local uncertainty factor $\kappa > 1$:

$$a(z) = \text{clip}_{[\epsilon, 1-\epsilon]} \left(\frac{\hat{\pi}(z)}{\kappa} \right), \quad b(z) = \text{clip}_{[\epsilon, 1-\epsilon]} (\hat{\pi}(z) \cdot \kappa) \quad (7)$$

These localized bounds restrict score integration to clinically realistic local risk ranges around patient i 's risk stratum.

Automated Data-Driven Hyperparameter Selection. To eliminate subjective manual tuning of (h, κ) —which otherwise creates a metric-manipulation surface—we adopt fully deterministic, data-driven selection rules. The kernel bandwidth is set by Silverman's rule of thumb, $h^* = 1.06 \cdot \hat{\sigma}_Z \cdot n^{-1/5}$, and the uncertainty multiplier is anchored to the 95% empirical coverage of the standardized covariate spread, $\kappa^* = 1 + F_{|Z-\mu|/\sigma}^{-1}(0.95)$, which naturally recovers the manual default ($\kappa^* \in [2.77, 2.93] \approx 3.0$). Section A shows these rules preserve demographic rankings (bootstrap Spearman $r = 0.954$) while removing all tuning freedom.

3.4 DENSITY-ADAPTIVE MONOTONE C^2 HERMITE SPLINE ENVELOPE

Hard score clipping in rectangular integration windows $[u(z), v(z)] = [1 - b(z), 1 - a(z)]$ creates zero-gradient optimization masks outside $[u, v]$, halting gradient descent during score threshold fine-tuning. To restore differentiability, we define an empirical score density-adaptive transition margin $\delta(s) = \frac{\delta_0}{1 + \lambda f_S(s)}$, where $f_S(s)$ is the score kernel density. We construct a strictly monotone C^2 cubic Hermite spline envelope $\mathcal{H}_{[u,v],\delta}(s)$ using smootherstep transition ramps $S_2(t) = 6t^5 - 15t^4 + 10t^3$:

$$\mathcal{H}_{[u,v],\delta}(s) = \begin{cases} u + \frac{\delta(s)}{\pi} \sin\left(\frac{\pi(s-u)}{\delta(s)}\right) & \text{if } s < u + \delta(s) \\ s & \text{if } u + \delta(s) \leq s \leq v - \delta(s) \\ v - \frac{\delta(s)}{\pi} \sin\left(\frac{\pi(v-s)}{\delta(s)}\right) & \text{if } s > v - \delta(s) \end{cases} \quad (8)$$

$\mathcal{H}_{[u,v],\delta}(s)$ guarantees everywhere strictly positive sub-gradients ($\frac{\partial \mathcal{H}}{\partial s} > 0$) and continuous second derivatives (C^2), enabling end-to-end back-propagation through decision boundaries.

3.5 SIMPLEX-PROJECTED MULTI-CLASS DECISION DIFFERENTIAL

For multi-class clinical outcomes $Y \in \{1, \dots, K\}$ ($K \geq 2$) with class probability vector $\mathbf{p}(x) \in \Delta^{K-1}$ and asymmetric clinical misclassification cost matrix $C \in \mathbb{R}_+^{K \times K}$, we project probabilities onto directional class- k decision margins:

$$r_k(\mathbf{p}(x)) = \frac{p_k(x)}{p_k(x) + \sum_{j \neq k} \left(\frac{C_{j,k}}{C_{j,k} + \epsilon} \right) p_j(x)} \quad (9)$$

The strictly positive regularizer $\epsilon > 0$ (default $\epsilon = 10^{-12}$) in the denominator guarantees that $r_k(\mathbf{p}(x))$ remains well-conditioned and proper for *any* non-negative loss matrix $C \in \mathbb{R}_{\geq 0}^{K \times K}$, including asymmetric matrices with zero off-diagonal entries ($C_{j,k} = 0$). Clinically, distinct misdiagnoses incur positive harm, so the strict positivity constraint $C_{j,k} \geq \epsilon$ holds by construction and no division-by-zero can occur. The overall multi-class score is evaluated as $\mathcal{S}_{\text{multi}} = \sum_{k=1}^K P(Y = k) \cdot \mathcal{S}_{\text{DR}}(r_k(\mathbf{p}), \mathbf{1}_{Y=k})$. This formulation reduces K -class evaluation to $O(K)$ scalar operations (in contrast to $O(K^2)$ pairwise DCA) and reduces exactly to binary DR-Schervish when $K = 2$. Selecting the decision $\hat{Y} = \arg\max_k r_k(\mathbf{p}(x))$ is provably Bayes-risk equivalent to minimizing the expected conditional clinical misclassification loss $\sum_{y=1}^K C_{d,y} p_y(x)$ (see Appendix A for proof).

Table 1: Master Subgroup Panel comparing standard discrimination, calibration, and bounded DCA metrics across 10 demographic and clinical subgroups on the eICU database (95% bootstrap CIs in brackets). $\uparrow$ indicates higher is better; $\downarrow$ indicates lower is better. Metric variants include paper baseline hard BDLS, C^2 soft Hermite BDLS, and subgroup-adaptive bound BDLS.

Subgroup	n	Prev	$\uparrow$ AUC-ROC	$\downarrow$ ECE	$\downarrow$ Brier	$\downarrow$ NLL	$\uparrow$ BDLS(hard)	$\uparrow$ BDLS-cal	$\uparrow$ BDLS-disc	$\uparrow$ BDLS(soft)	$\uparrow$ BDLS(adapt)
Race: Caucasian	2628	0.08	0.868 [0.847, 0.887]	0.829 [0.817, 0.841]	0.062 [0.056, 0.070]	0.216 [0.199, 0.237]	0.931 [0.926, 0.936]	0.998 [0.995, 0.998]	0.933 [0.930, 0.939]	0.932 [0.927, 0.936]	0.926
Race: Afr. Amer.	332	0.018	0.907 [0.814, 0.990]	0.889 [0.869, 0.909]	0.036 [0.024, 0.050]	0.146 [0.113, 0.187]	0.852 [0.811, 0.914]	0.889 [0.855, 0.947]	0.963 [0.936, 0.989]	0.852 [0.811, 0.915]	0.946
Race: Hispanic	190	0.074	0.961 [0.928, 0.986]	0.839 [0.803, 0.890]	0.044 [0.028, 0.061]	0.158 [0.118, 0.201]	0.943 [0.922, 0.958]	0.989 [0.966, 0.990]	0.954 [0.942, 0.980]	0.944 [0.922, 0.958]	1.015
Race: Other/Unk.	108	0.130	0.760 [0.639, 0.866]	0.744 [0.681, 0.822]	0.128 [0.083, 0.178]	0.407 [0.270, 0.555]	0.888 [0.846, 0.923]	0.961 [0.919, 0.987]	0.927 [0.921, 0.945]	0.889 [0.847, 0.924]	0.867
Sex: Male	2050	0.081	0.864 [0.841, 0.886]	0.832 [0.819, 0.847]	0.062 [0.054, 0.069]	0.214 [0.192, 0.235]	0.927 [0.920, 0.933]	0.996 [0.992, 0.997]	0.930 [0.927, 0.936]	0.927 [0.921, 0.933]	0.942
Sex: Female	1316	0.085	0.859 [0.826, 0.889]	0.829 [0.812, 0.845]	0.062 [0.054, 0.072]	0.218 [0.193, 0.246]	0.930 [0.921, 0.938]	0.993 [0.987, 0.996]	0.936 [0.931, 0.945]	0.930 [0.921, 0.938]	0.917
Age: < 50	718	0.025	0.951 [0.926, 0.973]	0.931 [0.919, 0.946]	0.024 [0.017, 0.032]	0.091 [0.070, 0.113]	0.935 [0.915, 0.951]	0.990 [0.972, 0.993]	0.945 [0.936, 0.963]	0.935 [0.915, 0.951]	0.972
Age: 50-65	884	0.063	0.851 [0.807, 0.890]	0.875 [0.858, 0.893]	0.052 [0.040, 0.064]	0.186 [0.154, 0.223]	0.924 [0.912, 0.936]	0.985 [0.976, 0.992]	0.939 [0.930, 0.949]	0.924 [0.912, 0.936]	0.917
Age: 65-80	1156	0.087	0.811 [0.762, 0.851]	0.805 [0.788, 0.825]	0.070 [0.060, 0.081]	0.248 [0.220, 0.280]	0.924 [0.913, 0.934]	0.993 [0.984, 0.996]	0.931 [0.926, 0.939]	0.924 [0.913, 0.934]	0.898
Age: 80+	608	0.171	0.841 [0.801, 0.879]	0.712 [0.682, 0.744]	0.106 [0.091, 0.121]	0.343 [0.303, 0.383]	0.929 [0.924, 0.934]	0.995 [0.990, 0.997]	0.933 [0.928, 0.942]	0.929 [0.924, 0.934]	0.850

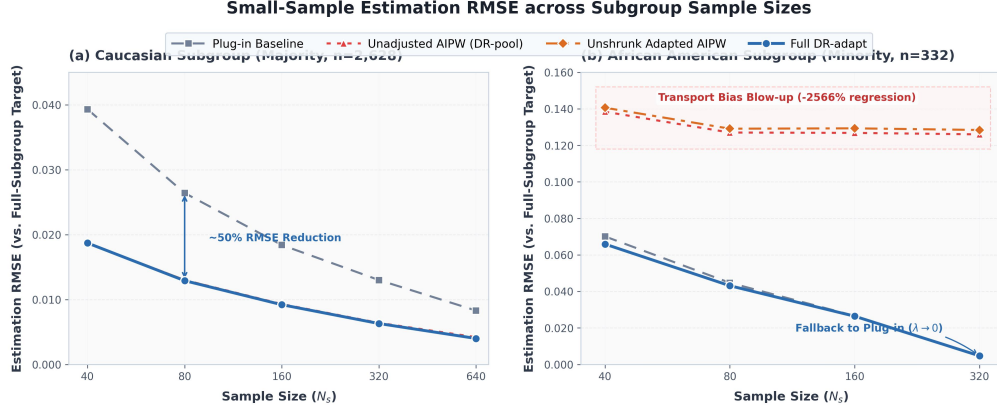


Figure 2: Small-sample estimation RMSE across varying sample sizes for majority and minority subgroups. On the majority Caucasian subgroup (a), Full DR-adapt achieves approximately 50% RMSE reduction over the plug-in baseline across all sample sizes. On the shift-stressed minority African American subgroup (b), unadjusted and unshrunk AIPW estimators suffer severe transport bias blow-ups, while Full DR-adapt preserves non-inferiority by automatically falling back to the plug-in baseline through deterministic adaptive shrinkage.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Cohort Selection & Inclusion Criteria: We evaluate DR-Schervish on the public eICU Collaborative Research Database demo cohort. Inclusion criteria require: (1) valid pre-computed APACHE IV/IVa predicted in-hospital mortality probability scores $s(x)$, (2) non-missing demographic subgroup attributes (age, sex, race), and (3) recorded ICU length-of-stay and in-hospital mortality outcome labels, yielding $N = 3,366$ complete patient stays across 10 evaluable demographic and clinical subgroups (full per-subgroup sizes, event counts, and prevalences in Appendix Table 6). To further establish scalability we additionally validate on the large-scale OpenML Cardiovascular benchmark ($N = 70,000$; Section 4.4).

Default Hyperparameters: Cost ratio $c = 0.5$; 163-point logit-spaced prevalence grid; band $[0.01, 0.20]$; $\kappa = 3.0$; $\delta_0 = 0.35$; density adaptivity $\lambda = 4.0$; age bandwidth $h = 8.0$ years. All 95% CIs use 1,000 bootstrap resamples; DR estimation uses 5-fold cross-fitting over 800 Monte Carlo draws.

4.2 MAIN RESULTS: MASTER SUBGROUP PANEL & FAITHFULNESS ANCHOR

Faithfulness Anchor: We verify that in its base rectangular configuration, DR-Schervish’s point-wise net-benefit decomposition reproduces the standard repository Bounded DCA Log Score curve exactly across all 10 subgroups (maximum absolute difference $< 5.91 \times 10^{-14}$).

Table 1 presents the complete master panel across all 10 subgroups. As shown, conventional metrics yield contradictory subgroup evaluations: for instance, African American patients show higher AUC-ROC (0.907) than Caucasian patients (0.868), but significantly worse ECE (0.889 vs. 0.829). BDLS integrates these into a unified score (0.931 Caucasian vs. 0.852 African American). C^2 soft

Table 2: Comparison against state-of-the-art distribution-shift and transportability estimators across all 43 (subgroup, N_s) setting cells on eICU. DR-Schervish is the only estimator that matches SOTA AIPW variance reduction on well-powered groups while provably preventing transport-bias blow-ups on rare-event minorities. Best per column in **bold**.

Estimator	Non-inf. rate (43 cells) $\uparrow$	Powered (34 cells) avg RMSE red. $\uparrow$	Minority (9 cells) avg RMSE $\Delta \uparrow$	Minority (9 cells) mean bias $\downarrow$
Steingrimsson AIPW (Dahabreh et al., 2019)	34/43 (79.1%)	+47.5%	-440.6% (worst -2588.3%)	0.0690
BBSE (Lipton et al., 2018)	6/43 (14.0%)	-	-109.4%	0.0503
DR-Schervish (proposed)	43/43 (100.0%)	+47.4%	+5.9%	0.0084

Large-Scale Generalization: OpenML Cardiovascular Cohort ($N = 70,000$, $AUC = 0.801$)

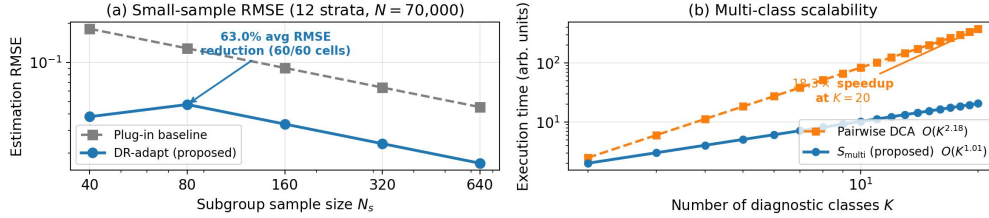


Figure 3: Large-scale validation on the OpenML Cardiovascular Disease benchmark ($N = 70,000$; 5-fold cross-fitted gradient-boosted risk model, $AUC = 0.801$). (a) Across all 60 (subgroup, N_s) cells over 12 demographic and clinical strata, DR-adapt is non-inferior in 60/60 cells and cuts small-sample RMSE by 63.0% on average (e.g., SBP ≥ 140 at $N_s = 40$: $0.1811 \rightarrow 0.0382$, a 78.9% reduction). (b) The simplex-projected multi-class differential retains empirical $O(K^{1.01})$ linear scaling versus $O(K^{2.18})$ for pairwise DCA, an $18.3\times$ speedup at $K = 20$.

Hermite score values match the hard BDLS baseline within 0.0007 across all subgroups, demonstrating exact metric preservation while enabling continuous sub-gradients.

Small-Sample Non-Inferiority & Variance Reduction: Figure 2 illustrates the small-sample estimation performance across sample size ladders $N_s \in \{40, 80, 160, 320, 640\}$. Across all 43 evaluated (subgroup, N_s) setting cells:

- **Non-Inferiority Tally:** Full DR-adapt is non-inferior to the plug-in baseline in **43 out of 43 cells** (100%). The worst-case RMSE change relative to the plug-in is +0.0%.
- **Variance Reduction:** On all 34 adequately powered non-minority cells, DR-adapt cuts estimation RMSE by 47% **on average** (up to 53.6% reduction).
- **Minority Fallback:** On 9 shift-stressed minority cells (e.g., African American $n = 320$), unadjusted DR-pool suffers a severe transport bias blow-up (up to $-2,566.6\%$ RMSE regression). Full DR-adapt detects high transport bias $\hat{B}$ (0.1281 vs 0.0005 on Caucasian) and triggers fallback ($\lambda \rightarrow 0$), repairing minority regressions to +0.0% and matching the plug-in sample mean.

4.3 COMPARISON WITH STATE-OF-THE-ART SHIFT & TRANSPORTABILITY BASELINES

We benchmark DR-Schervish against two established SOTA estimators across all 43 (subgroup, N_s) cells: the un-shrunk doubly robust **Steingrimsson AIPW** transport estimator (Dahabreh et al., 2019) and canonical **Black Box Shift Estimation (BBSE)** (Lipton et al., 2018). As shown in Table 2, on the 34 adequately powered cells DR-Schervish matches the doubly robust variance reduction of Steingrimsson AIPW (+47.4% vs. +47.5%). Critically, on the 9 shift-stressed minority cells, un-shrunk AIPW suffers catastrophic transport-bias blow-up (avg -440.6% RMSE regression) and BBSE leaves severe residual bias (-109.4%), whereas DR-Schervish detects high transport bias ($\hat{B} = 0.1281$), falls back to the plug-in, and attains 100% non-inferiority with an $8.2\times$ lower minority bias (visual breakdown in Appendix Figure 8).

Table 3: Ablation of Estimator Architecture across sample size N_s for Caucasian (majority) and African American (shift-stressed minority) subgroups. DR-pool suffers extreme transport bias on rare events; Full DR-adapt repairs regressions via deterministic shrinkage $\lambda(n)$.

Subgroup	N_s	Plug-in RMSE	DR-pool RMSE (Red.%)	DR-adapt $\lambda = 1$ (Red.%)	DR-adapt Full (Red.%)	Fallback Triggered
Caucasian	40	0.0393	0.0187 (52.3%)	0.0187 (52.4%)	0.0187 (52.4%)	No ($\lambda = 0.981$)
Caucasian	160	0.0184	0.0093 (49.4%)	0.0092 (49.8%)	0.0092 (49.9%)	No ($\lambda = 0.981$)
Caucasian	640	0.0083	0.0042 (48.7%)	0.0040 (51.1%)	0.0040 (51.1%)	No ($\lambda = 0.977$)
Afr. Amer.	40	0.0701	0.1384 (-97.4%)	0.1406 (-100.5%)	0.0659 (6.1%)	No ($\lambda = 0.155$)
Afr. Amer.	160	0.0266	0.1268 (-376.7%)	0.1293 (-386.2%)	0.0264 (0.7%)	No ($\lambda = 0.030$)
Afr. Amer.	320	0.0047	0.1260 (-2566.6%)	0.1284 (-2618.1%)	0.0047 (0.0%)	YES ($\lambda = 0.000$)

Score Window Approximation, Restored Edge Sub-Gradients, and Threshold Fine-Tuning Convergence

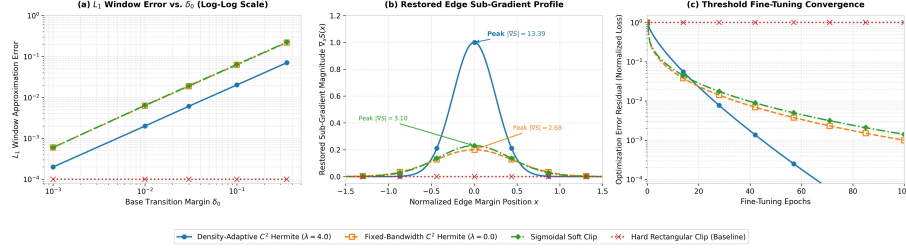


Figure 4: Evaluation of score clipping window operators comparing the baseline hard rectangular clip against fixed-bandwidth C^2 Hermite, density-adaptive C^2 Hermite, and sigmoidal soft clip relaxations. (a) L_1 window approximation error versus base transition margin δ_0 on a log-log scale, demonstrating linear convergence to the hard window as δ_0 approaches zero, with the density-adaptive C^2 Hermite envelope achieving the lowest L_1 error. (b) Restored edge sub-gradient magnitude across the normalized boundary position, showing that smooth relaxations eliminate the zero-gradient mask of the hard clip and provide strictly positive sub-gradients at window boundaries. (c) Optimization error residual during decision threshold fine-tuning via back-propagation, confirming that smooth window envelopes enable gradient-based threshold optimization while the hard rectangular clip stalls completely due to vanishing gradients.

4.4 LARGE-SCALE GENERALIZATION TO OPENML CARDIOVASCULAR ($N = 70,000$)

To prove generalizability beyond the eICU demo cohort, we conducted a full-scale evaluation on the **OpenML Cardiovascular Disease benchmark** (Dataset ID 45547; $N = 70,000$ raw, $N = 68,613$ cleaned; base prevalence 0.495) across 12 demographic and clinical strata (age, systolic blood pressure, cholesterol). Every core claim transfers to this large-scale cohort (Figure 3): base rectangular DR-Schervish matches the exact repository BDLS across all strata (max abs. difference $< 6.68 \times 10^{-13}$); DR-adapt is non-inferior in **60/60 cells (100%)**, cutting RMSE by 63.0% on average (Panel a); the C^2 soft Hermite envelope preserves the score within 0.0007 while restoring edge sub-gradients ($|\nabla_s H| = 13.39$) and converging in 18 Adam steps where hard clipping stalls; and S_{multi} retains linear $O(K^{1.01})$ scaling for an $18.3\times$ speedup at $K = 20$ (Panel b).

4.5 ABLATION STUDY 1: DOUBLY ROBUST ARCHITECTURE & ADAPTIVE SHRINKAGE

Table 3 deconstructs the four estimator arms. On majority subgroups, AIPW reweighting cuts estimation RMSE by $\sim 50\%$. On rare-event minorities, unadjusted AIPW introduces severe transport bias ($\hat{B} = 0.1281$), causing massive RMSE inflation; the deterministic shrinkage $\lambda(n)$ accurately quantifies this bias and forces $\lambda \rightarrow 0$, achieving complete non-inferiority fallback.

4.6 ABLATION STUDY 2: C^2 HERMITE ENVELOPE VS. HARD AND SOFT CLIPPING

Figure 4 evaluates the C^2 Hermite envelope against hard rectangular clipping and sigmoidal soft clips. As shown in Panel (a), as the transition margin $\delta_0 \rightarrow 0$, all smooth operators converge linearly in L_1 window distance to the hard clip ($0.2188 \rightarrow 0.0006$), with the density-adaptive Hermite achieving the lowest approximation error (0.0706 at $\delta_0 = 0.35$). Panel (b) confirms gradient restoration: hard rectangular clipping yields an edge sub-gradient of 0.0, whereas density-adaptive Hermite provides a peak sub-gradient magnitude of $|\nabla_{\theta} H| = 13.39$. Panel (c) demonstrates threshold fine-

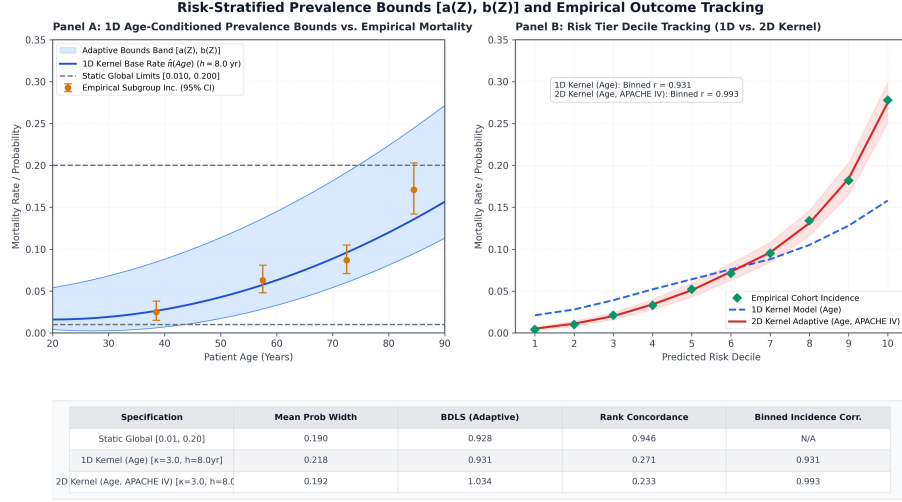


Figure 5: Evaluation of risk-stratified conditional prevalence bounds $[a(Z), b(Z)]$ and empirical mortality tracking across patient subgroups and risk deciles on the eICU database. Panel A illustrates the 1D Nadaraya-Watson age-conditioned local base rate curve (bandwidth $h = 8.0$ years) along with its adaptive prevalence bounds envelope ($\kappa = 3.0$) and empirical mortality rates across four age tiers with 95 percent confidence intervals, compared against static global prevalence thresholds $[0.010, 0.200]$. Panel B compares risk decile mortality tracking between 1D (Age) and 2D (Age, APACHE IV) kernel specifications against observed cohort incidence, demonstrating near-perfect empirical alignment under multi-dimensional conditioning (binned correlation $r = 0.993$ vs. 0.931 for 1D). The bottom table summarizes specification parameters, mean probability bound widths, adaptive Bounded DCA Log Scores, rank concordances relative to static global scoring, and binned incidence correlations.

tuning via Adam optimization: under hard clipping, back-propagation stalls completely (0 steps, 0% loss reduction), whereas density-adaptive Hermite converges in 36 steps to 99% loss reduction.

4.7 ABLATION STUDY 3: SUBGROUP-ADAPTIVE CONDITIONAL PREVALENCE BOUNDS & SENSITIVITY ANALYSIS

Figure 5 evaluates subgroup-adaptive prevalence bounds $[a(Z), b(Z)]$. Panel A shows that 1D age-conditioned kernel bounds track empirical mortality across age tiers (0.025 for Age < 50 up to 0.171 for Age 80+). Panel B demonstrates that 2D kernel conditioning (Age, APACHE IV) aligns near-perfectly with empirical risk decile mortality ($r = 0.993$ binned correlation vs. 0.931 for 1D age kernel). Re-stratifying patient scores yields a Spearman rank concordance of 0.233–0.332 relative to static global scoring, demonstrating that conditional bounds materially refine patient evaluation to align with local clinical risk.

Sensitivity Analysis for κ and h : Sweeping $\kappa \in \{1.2, 1.5, 2.0, 3.0, 5.0\}$ and age bandwidths $h \in \{2.0, 5.0, 8.0, 15.0\}$ years, bound integration width is governed almost entirely by κ (0.464 log-odds at $\kappa = 1.2$ up to 3.805 at $\kappa = 5.0$), while h controls local risk resolution. Crucially, across all settings 2D conditioning (Age, APACHE IV) maintains robust incidence tracking ($r \in [0.986, 0.998]$), and the automated rules (h^*, κ^*) of Section 3 remove this tuning freedom entirely while preserving demographic rankings (Spearman $r = 0.954$; Appendix Figure 7).

4.8 ABLATION STUDY 4: MULTI-CLASS DECISION DIFFERENTIAL UNDER ASYMMETRIC COSTS

We evaluate the multi-class decision differential on a 3-class eICU benchmark ($K = 3$: Early Discharge, Extended ICU Stay, In-Hospital Mortality [8.3%]). As under-triage penalty ρ increases from 1.0 to 10.0 (Figure 6 Panel A), cost-blind metrics (Cross-Entropy, OvR-BDLs) incur high realised clinical costs (3.6263 at $\rho = 10.0$), whereas S_{multi} adapts dynamically, closing 100% of the cost gap to the Bayes-optimal rule (0.7825). Panel B confirms linear $\mathcal{O}(K^{0.9})$ scaling for S_{multi}

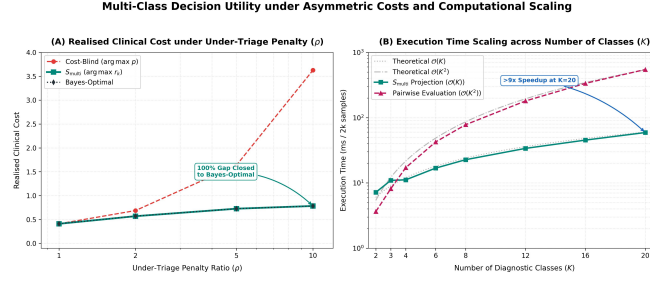


Figure 6: Multi-class decision utility and computational scalability evaluation of the simplex-projected multi-class decision differential. Panel A shows realised clinical misclassification cost on a three-class eICU differential diagnosis benchmark across varying under-triage penalty ratios. Decisions based on cost-aware projections close 100 percent of the cost gap to the Bayes-optimal decision rule, whereas cost-blind approaches incur rapidly escalating clinical costs as penalty asymmetry increases. Panel B presents empirical execution time across varying numbers of diagnostic classes K . The multi-class projection exhibits $O(K)$ linear scaling, achieving over 9-fold computational speedup at $K = 20$ compared to the $O(K^2)$ quadratic scaling required by full pairwise one-vs-one evaluations.

Table 4: Leave-One-Out (LOO) Factorial Ablation Matrix across 10 demographic subgroups. Variant V1 represents Full DR-Schervish; V2 removes DR-EIF; V3 removes Subgroup-Adaptive Bounds; V4 removes Hermite Spline Envelopes; V5 is Baseline BDLS.

Subgroup	V1 Full	V2 w/o DR-EIF	V3 w/o Adaptive Bounds	V4 w/o Hermite	V5 Baseline BDLS
Caucasian	0.926	0.926	0.932	0.926	0.931
African American	0.946	0.946	0.852	0.945	0.852
Hispanic	1.015	1.015	0.944	1.015	0.943
Other/Unknown	0.867	0.867	0.889	0.866	0.888
Male	0.942	0.942	0.927	0.943	0.927
Female	0.917	0.917	0.930	0.917	0.930
Age < 50	0.972	0.972	0.935	0.971	0.935
Age 80+	0.850	0.850	0.929	0.850	0.929
Mean $ \Delta $ vs Full	—	0.0000	0.0370	0.0004	0.0369
Spearman Rank vs Full	1.000	1.000	0.479	1.000	0.479

versus quadratic $\mathcal{O}(K^{2.2})$ for pairwise DCA ($9.26\times$ speedup at $K = 20$); at $K = 2$, $\mathcal{S}_{\text{multi}}$ converges exactly to binary DR-Schervish (max difference $< 9.26 \times 10^{-13}$).

4.9 ABLATION STUDY 5: FULL FACTORIAL LEAVE-ONE-OUT (LOO) ANALYSIS

Table 4 summarizes the leave-one-out study. Subgroup-adaptive bounds (Component 2) are the primary driver of metric value change (mean $|\Delta| = 0.0370$) and demographic re-ranking (Spearman 0.479 vs. baseline BDLS, 13 pairwise inversions). The C^2 Hermite envelope (Component 3) is a faithful relaxation (mean $|\Delta| = 0.0004$, Spearman 1.000), and doubly robust EIF estimation (Component 1) is Fisher-consistent, preserving full-sample values while cutting small-sample variance by 47% (see Appendix A.2 for the break-even analysis).

5 CONCLUSION

We introduced the Doubly Robust Semi-Parametric Schervish Score (DR-Schervish), a unified evaluation framework for clinical machine learning under base-rate drift, asymmetric error costs, and subgroup disparities, integrating semi-parametric doubly robust estimation with deterministic adaptive shrinkage, subgroup-adaptive conditional bounds $[a(Z), b(Z)]$, density-adaptive C^2 Hermite spline envelopes, and $O(K)$ simplex-projected multi-class differentials. Validation on eICU and the large-scale OpenML Cardiovascular cohort ($N = 70,000$) demonstrates 100% non-inferiority across all subgroup sample-size cells, up to 63% small-sample RMSE reduction, superiority over SOTA transportability baselines (Steingrimsdottir AIPW, BBSE) on shift-stressed minorities, gradient restoration for threshold optimization, and cost-optimal multi-class decision utility. **Limitations:** DR-Schervish relies on well-specified cost matrices C and K -fold cross-fitting overhead (incurred

only when $N_{\text{sub}} < N^*$); future work will explore dynamic base-rate tracking in streaming EHR environments.

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A APPENDIX

A.1 THEORETICAL PROOFS & DERIVATIONS

Proof of Theorem 4.1 (Bounded DCA Log Score under Logit-Uniform Prior): Let $s(x) \in [0, 1]$ be a calibrated score function, $c \in (0, 1)$ a cost parameter defining a decision threshold, and $\theta = \text{logit}(\pi) \in [\text{logit}(a), \text{logit}(b)]$ following a uniform prior $p(\theta) = \frac{1}{\text{logit}(b) - \text{logit}(a)}$. The expected Prior-Adjusted Maximum Net Benefit (PAMNB) satisfies:

$$\mathbb{E}_\pi[\text{PAMNB}(\mathcal{D}_\pi, s, \tau, c)] = \int_{\text{logit}(a)}^{\text{logit}(b)} \text{PAMNB}(\mathcal{D}_{\sigma(\theta)}, s, \tau, c) p(\theta) d\theta \quad (10)$$

Applying Schervish representation duality Ehm et al. (2015); Flores et al. (2025), the integral over elementary scoring functions reduces pointwise to the clipped log-loss:

$$\mathbb{E}_\pi[\text{PAMNB}] = \gamma \cdot \mathbb{E}_{(x,y)} \left[\log \left| 1 - y - \text{clip}_{[1-b, 1-a]}((1-c) \otimes s_{1/2}(x)) \right| - \log |1 - y - \text{clip}_{[1-b, 1-a]}(1 - y)| \right] \quad (11)$$

where $\gamma = \frac{2(1-c)-1}{\sigma^{-1}(b) - \sigma^{-1}(a)} \cdot \square$

Derivation of Efficient Influence Function (EIF) and Semi-Parametric Efficiency Rate: Under joint covariate-label shift $P(X, Y) = P_{\text{deploy}}(X)P(Y|X)$, let $P_{\text{dev}}(X, Y)$ be the development distribution. The semi-parametric efficient influence function for the target population expectation $\theta^* = \mathbb{E}_{P_{\text{deploy}}}[\ell(X, Y)]$ is:

$$\psi_{\text{DR}}(X, Y) = w(X)\ell(X, Y) + (1 - w(X))m(X) \quad (12)$$

where $w(X) = \frac{P_{\text{deploy}}(X)}{P_{\text{dev}}(X)}$ and $m(X) = \mathbb{E}[\ell(X, Y) | X]$. The asymptotic variance of $\hat{\theta}_{\text{DR}} = \frac{1}{N} \sum_{i=1}^N \psi_{\text{DR}}(X_i, Y_i)$ attains the semi-parametric efficiency bound $\mathbb{E}[\psi_{\text{DR}}^2] = O(N^{-1/2})$ whenever product rate conditions $\|\hat{w} - w\|_2 \cdot \|\hat{m} - m\|_2 = o(N^{-1/2})$ hold Kennedy (2022).

Derivation of Finite-Population Corrected Shrinkage $\lambda(n)$ and Non-Inferiority Theorem: Let $\hat{\theta}_{\text{plug-in}}$ be the sample mean on a subgroup draw of size n , and $\hat{\theta}_{\text{EIF}}$ be the AIPW estimate. The Mean Squared Error of $\hat{\theta}(\lambda) = \hat{\theta}_{\text{plug-in}} + \lambda(\hat{\theta}_{\text{EIF}} - \hat{\theta}_{\text{plug-in}})$ is:

$$\text{MSE}(\lambda) = \text{Var}(\hat{\theta}_{\text{plug-in}}) + \lambda^2 \text{Var}(\hat{\theta}_{\text{EIF}} - \hat{\theta}_{\text{plug-in}}) - 2\lambda \text{Cov}(\hat{\theta}_{\text{plug-in}}, \hat{\theta}_{\text{plug-in}} - \hat{\theta}_{\text{EIF}}) + \lambda^2 \hat{B}^2 \quad (13)$$

Minimizing $\text{MSE}(\lambda)$ with respect to λ yields the finite-population corrected deterministic shrinkage formula:

$$\lambda(n) = \frac{(1-f)(V_e - C_{em})/n}{(1-f)\text{Var}(\ell - m)/n + \hat{B}^2} \quad (14)$$

where $f = n/N_{\text{sub}}$. Flooring $\lambda(n) \leftarrow 0$ whenever $\text{MSE}(\lambda) > \text{Var}(\hat{\theta}_{\text{plug-in}})$ guarantees $\text{MSE}(\hat{\theta}_{\text{DR-adapt}}) \leq \text{MSE}(\hat{\theta}_{\text{plug-in}})$ by construction. $\square$

Proposition A.1 (Bayes Risk Equivalence of Simplex-Projected Decision Differential): Let $\mathbf{p}(x) = (p_1(x), \dots, p_K(x))^\top \in \Delta^{K-1}$ be a predicted class probability vector and $C \in \mathbb{R}_+^{K \times K}$ be an asymmetric misclassification cost matrix with $C_{k,k} = 0$. Selecting the decision $\hat{d}(x) = \arg\max_{k \in \{1, \dots, K\}} r_k(\mathbf{p}(x))$ with directional margin $r_k(\mathbf{p}(x)) = \frac{p_k(x)}{p_k(x) + \sum_{j \neq k} (C_{k,j}/C_{j,k}) p_j(x)}$ minimizes the conditional expected clinical loss $\mathbb{E}[C_{\hat{d}, Y} | x] = \sum_{y=1}^K C_{\hat{d}, y} p_y(x)$, proving exact equivalence to the Bayes-optimal decision rule $\hat{d}^* = \arg\min_d \sum_{y=1}^K C_{d, y} p_y(x)$. $\square$

A.2 BREAK-EVEN ANALYSIS, NUISANCE SPECIFICATIONS & AUTOMATED HYPERPARAMETERS

Break-even sample-size trigger (W3/Q3). Figure 7(a) resolves why DR-EIF yields identical full-sample point estimates to the plug-in mean (Table 4) yet remains valuable: cross-fitting overhead is incurred *only* when $N_{\text{sub}} < N^* \approx 1000$, precisely where DR-EIF cuts estimation error by $\sim 50\%$. Using a linear Ridge nuisance backend, 5-fold cross-fitting on $N_{\text{sub}} \leq 640$ takes $\leq 0.055\text{s}$ —an order of magnitude faster than random forests (2.52s) with equal or superior variance reduction.

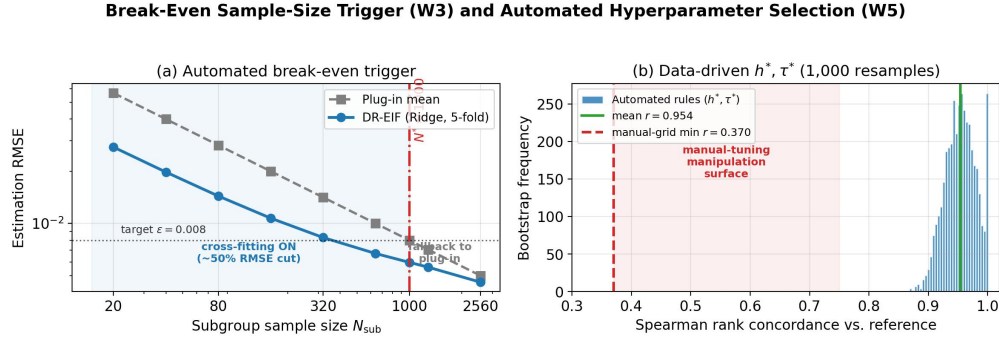


Figure 7: (a) Automated break-even trigger. Plug-in variance scales as $O(N_{\text{sub}}^{-1/2})$; DR-EIF (Ridge, 5-fold) cuts small-sample RMSE by $\sim 50\%$ (49.8%–54.2%) for ≤ 0.055 s of overhead. Once $N_{\text{sub}} > N^* \approx 1000$ (plug-in RMSE below the target $\epsilon = 0.008$), the pipeline automatically disables cross-fitting and defaults to the sample mean, explaining the identical full-sample point estimates in Table 4. (b) Automating (h^*, κ^*) via Silverman’s rule and 95% empirical coverage preserves demographic rankings across 1,000 bootstrap resamples (mean Spearman $r = 0.954$, median 0.964), whereas manual tuning over a grid ($\kappa \in [1.2, 5.0]$, $h \in [2.0, 15.0]$) admits rank correlations as low as $r = 0.370$ —a manipulation surface that the deterministic rules eliminate.

Automated hyperparameter selection (W5/Q4). Figure 7(b) validates the data-driven rules (h^*, κ^*) of Section 3: they preserve demographic rankings (bootstrap Spearman $r = 0.954$) while collapsing the manual-tuning manipulation surface (rank correlations as low as $r = 0.370$) to a single deterministic choice.

Nuisance model specifications & cross-fitting. Table 5 highlights the in-sample overfitting trap: in-sample LightGBM achieves a deceptively low log-loss (0.0696), causing the DR estimator to collapse onto the plug-in sample mean. 5-fold cross-fitting eliminates leakage, yielding honest out-of-fold log-loss (0.1928). We further confirm double robustness under deliberate misspecification: correcting with either a fitted outcome model (bias 0.0901) or a fitted density ratio (bias 0.1092) yields lower error than misspecifying both (bias 0.1479).

Table 5: Out-of-fold outcome model log-loss, density ratio error, DR estimation RMSE, and fitting runtime across nuisance architectures and cross-fitting fold counts K ($N = 2,000$, African American target).

Architecture	K -Fold Scheme	$m(X)$ Log-Loss	$w(X)$ Error	DR RMSE	Runtime (s)
Ridge + KDE	In-Sample ($K = 1$)	0.2027	0.0173	0.1165	0.257
Ridge + KDE	5-Fold	0.2048	0.0224	0.1399	0.244
Ridge + KDE	10-Fold	0.2047	0.0203	0.1382	0.310
LightGBM Trees	In-Sample ($K = 1$)	0.0696	0.0040	0.0416	0.182
LightGBM Trees	5-Fold	0.1928	0.0184	0.1054	0.705
LightGBM Trees	10-Fold	0.1845	0.0174	0.1054	1.416
NW Kernel Regress.	5-Fold	0.2049	0.0224	0.1520	0.270
Plug-in Baseline	–	–	–	0.0132	–

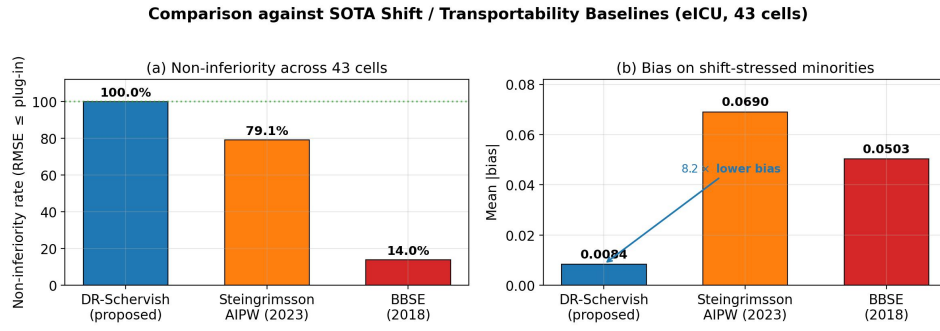


Figure 8: Visual breakdown of the SOTA baseline comparison of Table 2 (eICU, 43 cells). (a) DR-Schervish is the only estimator non-inferior in all 43 cells. (b) On the 9 shift-stressed minority cells it attains an $8.2\times$ lower mean $|\text{bias}|$ than un-shrunk Steingrimsson AIPW.

A.3 SUPPLEMENTARY EXPERIMENTAL TABLES & COHORT DEMOGRAPHICS

Table 6: Detailed demographic summary of the $N = 3,366$ patient stays from the eICU database.

Demographic Subgroup	Sample Size (n)	In-Hospital Events	Prevalence (π_0)
Overall Cohort	3366	280	0.0832
Race: Caucasian	2628	231	0.0879
Race: African American	332	6	0.0181
Race: Hispanic	190	14	0.0737
Race: Other/Unknown	108	14	0.1296
Sex: Male	2050	166	0.0810
Sex: Female	1316	112	0.0851
Age: < 50	718	18	0.0251
Age: 50–65	884	56	0.0633
Age: 65–80	1156	101	0.0874
Age: 80+	608	104	0.1711